

When Does Aggregating Explanations Work?

BloodMNIST Results

Jinhua Xu, Davide Anguita, Fabio Roli, Jing Yuan, and Luca Oneto

This report consolidates the released **BloodMNIST** experiment results. Tables are grouped by model, experimental protocol, and aggregation setting. Metric directions and robustness definitions are stated in every table header and caption.

Protocol	Model	Settings
Full-matrix protocol	ViT-B/16	NAIVE, IND, NOISE-S, NOISE-K

Table 1: NAIVE results for BLOODMNIST with ViT-B/16 (Full-matrix protocol). The methods are the best individual explanation, Simple Averaging, Borda, Kemeny–Young, RRF, and Schulze. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 3421$ samples.

Method	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
	F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Best Individual	0.306	0.377	0.317	0.386	0.082	0.075	0.085	0.065	0.064	0.054	0.067	0.044	0.042	0.030	0.040	0.029	0.029	-0.062	0.030	-0.062
Simple Averaging	0.302	0.408	0.313	0.418	0.060	0.066	0.066	0.059	0.046	0.055	0.053	0.048	0.012	0.009	0.014	0.009	-0.004	-0.114	-0.004	-0.115
Borda	0.258	0.444	0.269	0.454	0.073	0.102	0.080	0.096	0.056	0.073	0.065	0.066	0.024	0.027	0.026	0.026	0.039	-0.096	0.040	-0.096
Kemeny–Young	0.279	0.431	0.289	0.440	0.088	0.097	0.095	0.091	0.064	0.075	0.072	0.067	0.034	0.037	0.037	0.037	0.038	-0.082	0.038	-0.082
RRF	0.291	0.417	0.301	0.426	0.078	0.089	0.084	0.082	0.061	0.072	0.068	0.064	0.023	0.026	0.025	0.026	0.029	-0.095	0.029	-0.096
Schulze	0.272	0.429	0.282	0.438	0.087	0.096	0.095	0.089	0.063	0.073	0.071	0.065	0.030	0.030	0.033	0.030	0.034	-0.098	0.036	-0.098

Table 2: IND results for BLOODMNIST with ViT-B/16 (Full-matrix protocol). The Best Individual row is followed by matched NAIVE and IND results for each aggregation method at $q = 11$. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized. The test split contains $n = 3421$ samples.

Method	Setting	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Best Individual	baseline	0.191	0.449	0.199	0.459	0.059	0.080	0.066	0.069	0.041	0.057	0.048	0.048	0.043	0.044	0.044	0.044	0.064	-0.023	0.064	-0.023
Simple Averaging	matched NAIVE	0.179	0.478	0.186	0.487	0.064	0.061	0.072	0.052	0.048	0.045	0.058	0.036	0.049	0.039	0.052	0.038	0.057	-0.018	0.058	-0.018
Simple Averaging	IND	0.180	0.462	0.187	0.471	0.059	0.075	0.069	0.065	0.049	0.046	0.058	0.036	0.048	0.036	0.051	0.036	0.050	-0.023	0.051	-0.023
Borda	matched NAIVE	0.181	0.483	0.188	0.493	0.072	0.081	0.081	0.071	0.061	0.064	0.070	0.055	0.053	0.056	0.056	0.055	0.066	-0.020	0.067	-0.020
Borda	IND	0.183	0.452	0.189	0.462	0.079	0.077	0.089	0.067	0.064	0.056	0.073	0.047	0.054	0.053	0.056	0.052	0.059	-0.023	0.060	-0.023
Kemeny–Young	matched NAIVE	0.181	0.502	0.189	0.512	0.070	0.060	0.077	0.050	0.057	0.050	0.065	0.041	0.054	0.045	0.056	0.045	0.060	-0.015	0.061	-0.015
Kemeny–Young	IND	0.189	0.464	0.196	0.474	0.069	0.069	0.077	0.059	0.056	0.060	0.063	0.051	0.057	0.049	0.060	0.049	0.047	-0.023	0.048	-0.024
RRF	matched NAIVE	0.181	0.490	0.189	0.500	0.073	0.068	0.081	0.058	0.059	0.062	0.066	0.053	0.055	0.052	0.057	0.051	0.061	-0.018	0.063	-0.018
RRF	IND	0.187	0.455	0.194	0.465	0.067	0.074	0.076	0.064	0.058	0.057	0.065	0.049	0.051	0.050	0.054	0.050	0.059	-0.022	0.060	-0.021
Schulze	matched NAIVE	0.183	0.495	0.190	0.505	0.068	0.071	0.077	0.061	0.057	0.058	0.066	0.049	0.052	0.051	0.054	0.050	0.060	-0.018	0.061	-0.018
Schulze	IND	0.190	0.451	0.196	0.461	0.068	0.070	0.079	0.061	0.056	0.049	0.066	0.041	0.050	0.051	0.053	0.050	0.056	-0.025	0.057	-0.025

Table 3: NOISE-S results for BloodMNIST with ViT-B/16 (Full-matrix protocol), using the Spearman-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	3	0.306	0.359	0.317	0.369	0.088	0.060	0.093	0.052	0.065	0.043	0.070	0.035	0.040	0.023	0.041	0.024	0.059	-0.094	0.059	-0.094
Borda	3	0.321	0.368	0.331	0.377	0.077	0.079	0.083	0.071	0.055	0.063	0.061	0.055	0.037	0.025	0.038	0.025	0.050	-0.065	0.049	-0.066
Kemeny–Young	3	0.311	0.369	0.322	0.378	0.077	0.070	0.081	0.058	0.059	0.052	0.064	0.043	0.034	0.027	0.034	0.026	0.038	-0.059	0.037	-0.060
RRF	3	0.321	0.361	0.331	0.370	0.083	0.070	0.088	0.061	0.061	0.056	0.067	0.047	0.039	0.025	0.040	0.025	0.051	-0.063	0.051	-0.063
Schulze	3	0.312	0.363	0.322	0.371	0.082	0.073	0.088	0.064	0.061	0.054	0.067	0.045	0.041	0.024	0.042	0.023	0.053	-0.059	0.053	-0.061

Table 4: NOISE-K results for BLOODMNIST with ViT-B/16 (Full-matrix protocol), using the Kendall-based rank-noise geometry. The q column gives the selected explanation-prefix length. All metrics use $p = 16$, $k = 20$, dataset-mean filling, $\sigma_g = 0.15$, $\pi_p = 0.05$, $\sigma_s = 0.15$, and $\epsilon = 2/255$. Robustness is signed direction-aware quality change; positive values denote improvement under perturbation and all R columns are maximized.

Method	q	Clean quality				Gaussian (g)				Salt-and-pepper (p)				Speckle (s)				Adversarial (a)			
		F $\uparrow$	$\bar{F}$ $\downarrow$	C $\uparrow$	$\bar{C}$ $\downarrow$	R _F ^g $\uparrow$	R _{$\bar{F}$} ^g $\uparrow$	R _C ^g $\uparrow$	R _{$\bar{C}$} ^g $\uparrow$	R _F ^p $\uparrow$	R _{$\bar{F}$} ^p $\uparrow$	R _C ^p $\uparrow$	R _{$\bar{C}$} ^p $\uparrow$	R _F ^s $\uparrow$	R _{$\bar{F}$} ^s $\uparrow$	R _C ^s $\uparrow$	R _{$\bar{C}$} ^s $\uparrow$	R _F ^a $\uparrow$	R _{$\bar{F}$} ^a $\uparrow$	R _C ^a $\uparrow$	R _{$\bar{C}$} ^a $\uparrow$
Simple Averaging	3	0.306	0.359	0.317	0.369	0.088	0.060	0.093	0.052	0.065	0.043	0.070	0.035	0.040	0.023	0.041	0.024	0.059	-0.094	0.059	-0.094
Borda	3	0.321	0.368	0.331	0.377	0.077	0.079	0.083	0.071	0.055	0.063	0.061	0.055	0.037	0.025	0.038	0.025	0.050	-0.065	0.049	-0.066
Kemeny–Young	3	0.311	0.369	0.322	0.378	0.077	0.070	0.081	0.058	0.059	0.052	0.064	0.043	0.034	0.027	0.034	0.026	0.038	-0.059	0.037	-0.060
RRF	3	0.321	0.361	0.331	0.370	0.083	0.070	0.088	0.061	0.061	0.056	0.067	0.047	0.039	0.025	0.040	0.025	0.051	-0.063	0.051	-0.063
Schulze	3	0.312	0.363	0.322	0.371	0.082	0.073	0.088	0.064	0.061	0.054	0.067	0.045	0.041	0.024	0.042	0.023	0.053	-0.059	0.053	-0.061